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Task 1:

```
#include<iostream>

using namespace std;

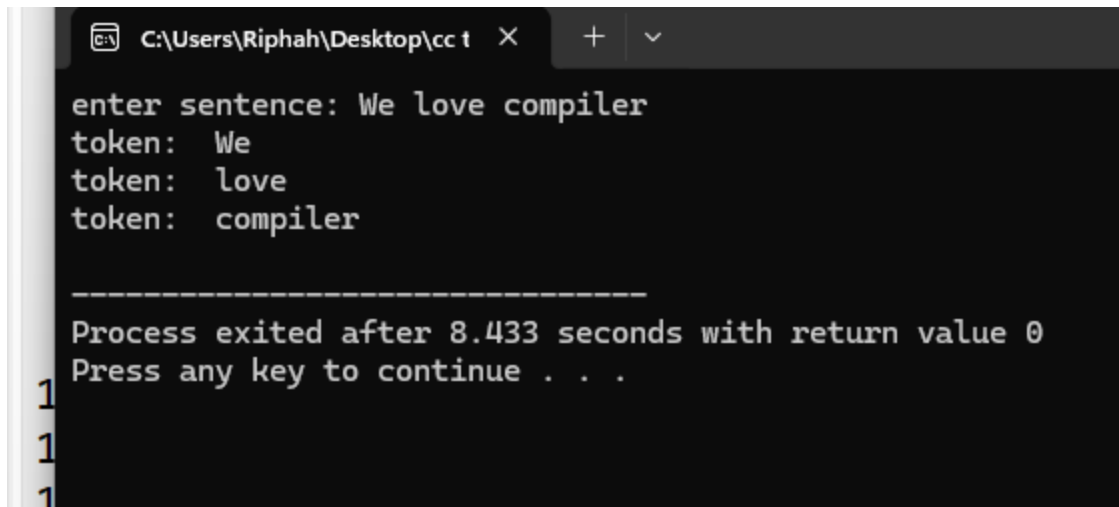
int main()
{
    string input;

    cout<<"enter sentence: ";
    getline(cin,input);

    string token=" ";

    for(int i=0;i<input.length();i++)
    {
        if (input[i] !=' ')
        {
            token+=input[i];
        }
        else{
            cout<<"token: "<<token<<endl;
            token=" ";
        }
    }

    cout<<"token: "<<token<<endl;
    return 0;
}
```



```
C:\Users\Riphah\Desktop\cc t X + v
enter sentence: We love compiler
token: We
token: love
token: compiler

-----
Process exited after 8.433 seconds with return value 0
Press any key to continue . . .
```

Task 2:

```
#include <iostream>

using namespace std;

int main()
{
    string input;
    cout<<"enter statement: \n";
    getline(cin,input);
    string token = "";
    for(int i=0;i<input.length();i++)
    {
        if(input[i]!=' ')
        {
            token+=input[i];
        }
        else{
            if(token=="int" || token=="float" || token=="string" || token=="char")
            cout<<token<<"KEYWORD\n";
        }
    }
}
```

```

else if (token == "+" || token == "=")
cout<<token<<":operator\n";
else if(token[0]>='0' && token[0]<='9')
cout<<token<<":NUMBER\n";
else if(token[0] == ';')
cout<<token<<": TERMINATOR\n";
else
cout<<token<<": IDENTIFIER\n ";
token="";
}
}

if(token=="int" || token=="float" || token=="string" || token=="char")
cout<<token<<"KEYWORD\n";
else if (token=="+" || token=="=")
cout<<token<<":operator\n";
else if(token[0]<='0' && token[0]<='9')
cout<<token<<":NUMBER\n";
else if(token[0] == ';')
cout<<token<<": TERMINATOR\n";
else {
cout<<token<<": IDENTIFIER\n ";
}
}
}

```

```
C:\Users\Riphah\Desktop\pra X + v
enter statement:
int total = sum + 10
intKEYWORD
total: IDENTIFIER
=:operator
sum: IDENTIFIER
+:operator
10: IDENTIFIER

-----
Process exited after 6.729 seconds with return value 0
Press any key to continue . . . |
```

Task 3:

```
#include <iostream>

using namespace std;

int main()
{
    string input;
    cout<<"enter statement: \n";
    getline(cin,input);
    string token = "";
    for(int i=0;i<input.length();i++)
    {
        if(input[i]!=' ')
        {
            token+=input[i];
        }
        else{
```

```

if(token=="int" || token=="float" || token=="string" || token=="char")
cout<<token<<"KEYWORD\n";
else if (token == "+" || token == "=")
cout<<token<<":operator\n";
else if(token[0]>='0' && token[0]<='9')
cout<<token<<":NUMBER\n";
else if(token[0] == ';')
cout<<token<<": TERMINATOR\n";
else if (token == "if" || token == "else if")
cout<<token<<": Conditional statement\n";
else if (token == "for" || token == "while")
cout<<token<<": loop statement\n";
else
cout<<token<<": IDENTIFIER\n ";
token="";
}
}

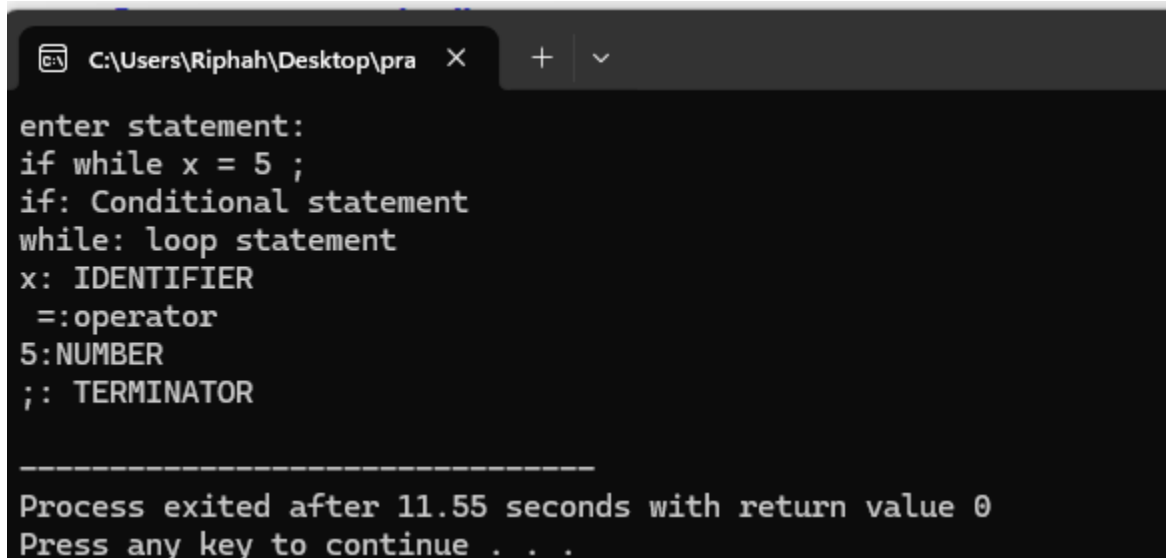
```

```

if(token=="int" || token=="float" || token=="string" || token=="char")
cout<<token<<"KEYWORD\n";
else if (token=="+" || token=="=")
cout<<token<<":operator\n";
else if(token[0]<='0' && token[0]<='9')
cout<<token<<":NUMBER\n";
else if(token[0] == ';')
cout<<token<<": TERMINATOR\n";
else if (token == "if" || token == "else if")
cout<<token<<": Conditional statement\n";
else if (token == "for" || token == "while")
cout<<token<<": loop statement\n";

```

```
        else {  
            cout<<token<<": IDENTIFIER\n";  
        }  
    }  
}
```



```
C:\Users\Riphah\Desktop\pra X + v  
enter statement:  
if while x = 5 ;  
if: Conditional statement  
while: loop statement  
x: IDENTIFIER  
=:operator  
5:NUMBER  
;: TERMINATOR  
  
-----  
Process exited after 11.55 seconds with return value 0  
Press any key to continue . . .
```